

For this first introduction to this math forum, I wanted to feature something that was unlikely to be known (it's very rarely taught in schools) but that was approachable by any level in mathematics, and still featured some very interesting results.

I will try to keep a more *motivated* approach; instead of just throwing formulas or showing solutions like I would in overkill math, in this section the objective will be to get to results from a motivated point of view, by exploring ideas until arrival to a result.

So let it begin.

1 A curious discovery

What I have picked for our first dive into mathematics is *modular arithmetic*. The name is daunting at first, but the concept itself will be simple to grasp.

What we'll consider here is going down to the very basics. What is the most basic mathematics one can think about?

The easy answer would be addition, subtraction, multiplication and division. And that's exactly what we'll think about for now.

What we'll consider for now is the notion of division. We've all learned the good old method of dividend, divisor, quotient and remainder. That is, if we have a dividend (the number we divide) a , and our divisor b , we get something like

$$a = bq + r$$

For some q (called the quotient) and an r that is between 0 and $b - 1$. For example, when dividing 101 over 7, one would get

$$101 = 7 \cdot 14 + 3$$

Which gives us a quotient of 14 and a remainder of 3. We can take a couple more examples:

$$38 = 7 \cdot 5 + 3$$

$$43 = 7 \cdot 6 + 1$$

If we were to play around with these examples, we can find an unsurprising but interesting pattern:

$$101 - 38 = (7 \cdot 14 + 3) - (7 \cdot 5 + 3) = 63 = 7 \cdot 9 + 0$$

$$101 - 43 = (7 \cdot 14 + 3) - (7 \cdot 6 + 1) = 7 \cdot 8 + 2$$

It would seem like the remainders act on their own, without really being interested in how the rest behaves. And that makes a lot of sense, since they literally act separately; they are the part that isn't a multiple of 7. If we consider multiplication:

$$101 \cdot 43 = (7 \cdot 14 + 3)(7 \cdot 6 + 1) = 4343 = 7 \cdot 620 + 3$$

Again we find that the remainders act independently.

Now, imagine you're a 7; you take in numbers, break them up as best as possible, and leave some remaining part. For you, this extra bit isn't really important unless it's large enough to matter.

So, in a sense, for you, classifying numbers by their remaining part is the most natural thing there is!

So let's consider how 7 might classify everything. We'll imagine that 7 just took = and added one stroke $\equiv$ to denote this equivalence relation of remainders.

So from what we know, for example, we could say that $101 \equiv 38$, since they both have remainder 3. It's also clear that $1 \equiv 43$, since both have remainder 1. Like this, we can essentially relate every number from the point of view of a 7.

Imagine then that you have some numbers x, y and that they in fact verify $x \equiv y$; that is, they have the same remainder when divided by 7. The great thing is that using what we've discovered from remainders, we know how their sum, difference, and product work:

$$2x \equiv x + y \equiv 2y$$

$$x - y \equiv 0$$

$$x^2 \equiv xy \equiv y^2$$

This looks nice and all, but how is it actually useful?

1.1 A first application

Let's say that as a 7 you're trying to understand the behaviour of numbers of the form

$$15^n - 8^n$$

Where n is some positive integer. How might you decide if this has a remainder? (*Try this!*)

We'll try some cases for now:

$$15^1 - 8^1 = 7 \equiv 0$$

$$15^2 - 8^2 = 225 - 64 = 161 \equiv 0$$

So it'd seem as if we always get a remainder of 0.

Well, we do know some basic information about 15 and 8. In particular, we know that $8 \equiv 1$ and $15 \equiv 1$. But then for us this is really the same as

$$(1)^n - (1)^n \equiv 0$$

So we have that the remainder is 0 at every value of n .

It's worth to make a pause here. In a normal math class, one could find a problem like this maybe as a pretty tough induction problem, or not even get shown such a problem because of its difficulty, as you don't really have formulas to solve it. But really, as we've seen, it's essentially a one line proof!

This is the power of *modular arithmetic*. What we've worked on now is exactly this at work; a mathematician would call $\equiv$ *congruence* and point out that we were just working *modulo* 7.

For now, we'll leave a couple of example problems to work on modulus of different numbers:

1. Consider all numbers of the form $7^n - 3^n$. Are they divisible by 4?
2. It is well known that the sum of the first n squares $(1^2 + 2^2 + \dots + n^2)$ is $\frac{n(n+1)(2n+1)}{6}$. Determine whether or not this fraction always return an integer value (that is, $n(n+1)(2n+1) \equiv 0$ modulo 6 for each n).
3. Consider what values you can have for x^2 modulo 4. Using that information, determine if $x^2 + 4x + 6$ can ever be a perfect square.
4. We know that all prime numbers larger than 2 are odd. Consider what remainders they might leave modulo 4. Using this, determine if a number x such that $x \equiv 3$ modulo 4 necessarily has to be a multiple of some prime number p such that $p \equiv 3$ modulo 4. Argue if the same can be said about $x \equiv 1$ and a prime number $p \equiv 1$.

The power of modular arithmetic resides in the fact that it reduces everything to a couple of cases to treat, as now we're working with numbers only smaller than our modulo.

To show off just a small last part of modular arithmetic, we'll observe its utility with *divisibility rules*.

Some of us might have heard of a trick to divide by 3: if the sum of digits of the number is a multiple of 3, the number itself also is. For example, with

1278, checking directly its divisibility by 3 would take a while; however, since $1 + 2 + 7 + 8 = 18 = 3 \cdot 8$ is a multiple of 3, we know 1278 is too.

What most people do not know is *why* this is the case. But with modular arithmetic, this is a very natural fact!

Most of us know how to consider the decimal expansion of a number. For example, again with 1278:

$$1278 = 1 \cdot 10^3 + 2 \cdot 10^2 + 7 \cdot 10^1 + 8$$

But now consider this expansion modulo 3. We know that $10 \equiv 1$, so we really get this reduced to:

$$1278 \equiv 1 \cdot 1^3 + 2 \cdot 1^2 + 7 \cdot 1^1 + 8 \equiv 1 + 2 + 7 + 8$$

So we see that 1278 is congruent to the sum of its digits modulo 3! Thus, if 1278 was divisible by 3, we'd have:

$$1278 \equiv 0 \equiv 1 + 2 + 7 + 8$$

So we find the divisibility trick that we already knew! In fact, it's not hard to see why it should work for 9 as well. But what we have here is a nice way to generate divisibility tricks for any prime number we want! For example, there is a much less known rule for divisibility over 11. If we consider some number, say 1298, and we want to know if it is divisible by 11, we may take its decimal expansion:

$$1 \cdot 10^3 + 2 \cdot 10^2 + 9 \cdot 10 + 8$$

But we know that $10 \equiv -1$ modulo 11. What does it mean to be congruent to a negative number? Well, in theory, we could just say $-1 = 11 \cdot (-1) + 10$, so in a way -1 is really congruent to 10 modulo 11. What this is useful for is that we know how the powers of -1 act: they give -1 for odd powers, and 1 for even powers. Thus, we get an alternating sum:

$$1 \cdot 10^3 + 2 \cdot 10^2 + 9 \cdot 10 + 8 \equiv 1(-1) + 2(1) + 9(-1) + 8 \equiv 0$$

So we get that 1298 is in fact divisible by 11. What this alternating sum shows is that *if the sum of the even digits minus the sum of the odd digits is a multiple of 11, the number is a multiple of 11*. This is the much less known but equally effective rule for divisibility by 11.

For our big finale, we consider the last few digits of a number. For example, for 1298, the last 2 digits are 98. How can we encode this in modular arithmetic? (*Can you think of it?*)

Well, we know that for 100, the digits above the third will all be 0 (*why?*), so we know

$$1298 \equiv 98$$

Modulo 100. Thus, if we consider 10^n as our modulus, we extract the last n digits of a number. We now apply this to the following problem, with which we end this section.

A-4. Define a sequence a_i by $a_1 = 3$ and $a_{i+1} = 3^{a_i}$ for $i \geq 1$. Which integers between 00 and 99 inclusive occur as the last two digits in the decimal expansion of infinitely many a_i ?

Now this problem is *intimidating*. Instead of just using some cases, it defines a series (which looks like $3, 3^3, 3^{3^3}, 3^{3^{3^3}}$, etc) and tells us to know what digits it end on *infinitely many* times.

But for now let's just play around with it for a bit. For example, since we know we want the last 2 digits, we can consider it all modulo 100 to simplify. The key observation here is that trying out a few powers of 3, we find:

$$3, 9, 27, 81, 243, \dots$$

Which doesn't seem to lead anywhere; however, a hidden idea is underlying here. If we take specifically 243, considering it modulo 100 we have 43. Furthermore, the square of it is $43^2 = 1849 \equiv 49$. But what we find with 49 is that $49^2 = 2401 \equiv 1$.

The utility of this fact might not be immediately obvious, but this is huge. It tells us that $3^{20} \equiv 1$ modulo 100, which means that wherever in our sequence (remember, of the form $3, 3^3, 3^{3^3}, 3^{3^{3^3}}$) we can subtract 20 from the exponent, as it's the same as multiplying by 1. That is:

$$3^{3^{3^{\cdots}}} \equiv 3^{3^{3^{\cdots}} - 20} \cdot 3^{20}$$

And because 3^{20} is 1, we can ignore it. Essentially, what we have is that *we can treat the exponent modulo 20*.

But the exponent has exactly the same form $3, 3^3, 3^{3^3}, \dots$! So we might try to find another power of 3 that gives us 1 modulo 20. We find our candidate in $3^4 = 81 \equiv 1$.

So yet again, we can treat the second exponent modulo 4. If we consider powers of 3 modulo 4, we get $3^2 \equiv 1$. Finally, modulo 2 every power is always reduced to 1.

So how was this whole journey useful? Well, what we find is that since the fourth exponent (treated modulo 2) is *always* 1, the third exponent is always 3 (except at the second term, 3^3); then the second exponent is always $3^3 = 27 \equiv 7$ modulo 20 (except at the first two terms) and thus the whole thing is *always* 3^7 modulo 100 from the third term.

Thus, considering only the first three terms modulo 100:

$$3, 27, 87$$

We know, for a fact, that *every term* from the third will be 87 modulo 100. Thus, the only number between 00 and 99 that appears infinitely many times is 87.

And with this, we close this section. But, just as a comment, this last problem was extracted from the Putnam competition; a mathematics competition known worldwide for its difficulty (although, admittedly, it is an old edition; 1985). Still, it's something to be proud of!

2 A glaring flaw

In the previous section, we introduced the notions of *modular arithmetic*; and we found methods to deal with divisibility, perfect squares (and perfect powers in general, by extension), a way to find information about the prime factors of a number, and a way to obtain information about the ending digits of a number.

It is clear, I hope, how useful and flexible modular arithmetic is, since it allows us to deal with a variety of problems using nothing more than basic divisibility ideas. During our exploration and construction of modular arithmetic, we built the following table:

$$\begin{aligned}2x &\equiv x + y \equiv 2y \\ x - y &\equiv 0 \\ x^2 &\equiv xy \equiv y^2\end{aligned}$$

To treat the sum, difference, and product of two numbers $x \equiv y$. It is clear that we're lacking the last of the four big operations: division.

We could want to just be able to use division directly, but that breaks down quickly. For example, if we considered $x \equiv y \equiv 1$ modulo 4 (that is, $x = 4a + 1$ and $y = 4b + 1$), and imagine what their division should be like, modulo 4 we'd get $\frac{x}{y} \equiv \frac{1}{1} \equiv 1$; but $\frac{4a+1}{4b+1}$ could be anything, not even an integer.

So how can we use division then? Is all hope lost? Well, not necessarily. In situations like this, it is sometimes helpful to try to look at it more abstractly. Division isn't exactly an operation by itself, it's more like undoing multiplication (just like subtracting undoes addition and roots undo powers). If we think about dividing by 2, what we're really doing is finding the number that, multiplied by 2, will give us our input. To find it, we multiply by 0.5. So, essentially, a division is still a multiplication (*dividing* by 2 is *multiplying* by 0.5), just the other way around. How can we precisely note down this idea of undoing?

The Putnam problem right at the end of the last section might give us an idea (new math is found by trying to solve hard problems!). In there, we used the fact that $3^{20} \equiv 1$, so we could *cancel it out*. This idea is nice. Since multiplying by 1 does nothing, maybe we can undo multiplication by just converting it to 1.

Notice our example of 0.5 verifies this, since $2 \cdot 0.5 = 1$. The core idea is that if we have a product ab and we desire to extract b , we multiply it by some a' such that $a'a = 1$, so we get $a' \cdot (ab) = (a'a)b = 1 \cdot b = b$.

So we have a solution! Maybe $\frac{x}{y}$ doesn't translate exactly to $\frac{4a+1}{4b+1}$, but we do have an idea of how $\frac{x}{y}$ *should act like*.

An example could be the one just shown: 3^{19} is the multiplicative inverse of 3 modulo 100.

But we are missing one idea here. Can every number have such an a' ? It is clear, for example, that 0 doesn't. So is there any other numbers that don't? And if so, why?

Before treating these questions, we need to give a name to this a' . Since it undoes multiplication, it's not a far stretch to call it the *multiplicative inverse* of a . Now, we may ask more directly: what numbers have multiplicative inverses, and why?

Consider, for example, modulo 4. If we consider 1 and 3, they are their own multiplicative inverses (it is clear that 1 always has a multiplicative inverse, itself; and since $3 \equiv -1$, it's also easy to see why it should have a multiplicative inverse). However, when looking for a multiplicative inverse for 2, we're stuck. No matter what number we multiply it by, we get a number different from 1. Why is this?

Well, imagine $2a \equiv 1$. We know exactly what this translates to: $2a = 4x + 1$ for some x . But this should raise a red flag immediately. We have an even number on the left, and an odd number on the right! This is clearly wrong.

So what do we get from this? Well, it seems like if the modulus (in this case, 4) and the number (2) are of the same "type" (even) then we cannot have a multiplicative inverse. This notion of "type" is clear: they are both multiples of 2. So if the modulus and the number can both be divided by some number that is bigger than 1, then the number can't have a multiplicative inverse. But this is just their greatest common divisor!

Recall that the greatest common divisor of two numbers a, b (written $\gcd(a, b)$ or even just (a, b)) is just the biggest number that divides both a and b . So what we're saying is that if the number and the modulus have greatest common divisor bigger than 1, then the number can't have a multiplicative inverse. Numbers that do have multiplicative inverses would have to be sharing no factors with the modulus.

These kinds of numbers such that $\gcd(a, b) = 1$ (it should be clear why 1 is the lowest gcd possible) are very special. Imagine that you're a in this situation. For you, is there really any difference between b and any prime number? For you, they both are just numbers that you share nothing with, so b may as well be prime. That's why mathematicians call these pairs of numbers *coprime*; they just act like prime numbers to each other.

And now we have the full potential of modular arithmetic. If we write our table again

$$\begin{aligned} 2x &\equiv x + y \equiv 2y \\ x - y &\equiv 0 \\ x^2 &\equiv xy \equiv y^2 \\ \frac{x}{y} &\equiv 1 \text{ whenever } \gcd(x, n) = 1 \end{aligned}$$

Where n is the modulus. It's a bit different from normal division, but this extension is a gigantic tool we can use.

You may have heard prime numbers are a great object in mathematics, but most don't get to see why this is the case. Well, here's one of many bits that should make us love prime numbers: they are coprime with everything! Not counting 0, every number modulo a prime number p always has a multiplicative inverse. For example, modulo 7:

$$\begin{array}{c|c} 1 & 1 \\ 2 & 4 \\ 3 & 5 \\ 6 & 6 \end{array}$$

How are multiplicative inverses useful? Well, let's try out some of their properties and play around with them.

For example, an interesting idea is that now we can multiply all numbers together and we'll get cancellations everywhere:

$$1 \cdot 2 \cdot 3 \cdots (p-2) \cdot (p-1)$$

Since all of these have their multiplicative inverse *except 1 and -1*, we get that this product is just reduced to

$$1 \cdot 2 \cdot 3 \cdots (p-2) \cdot (p-1) \equiv -1$$

Notice that the left side is just a factorial, $(p-1)!$. What we also know is that if our modulus is not a prime, say n , then $(n-1)!$ shouldn't be -1, since then every number would be coprime with it (*Why?*) and that's just the definition of a prime number.

So we've devised a test for checking if a number is prime, and it involves a factorial, which we didn't know how to work with before! This is a pretty strong result, don't you think? Strong enough, in fact, that it has a name: *Wilson's Theorem*.

We still have one more idea to work with, but we take a little break here and get comfortable with the idea of multiplicative inverses with a couple practice problems. A hint is associated to each of them, available at the end of the section.

1. Find the multiplicative inverse of each nonzero number modulo 5.
2. Check that multiplicative inverses add just like fractions; that is, check that if b, d are numbers coprime with n , then modulo n we have $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$.
3. Check that if a, b are integers and $g = \gcd(a, b)$, then g divides $ax + by$ for all integers x, y .
4. Show that if a number a is coprime with n , then there are no two elements b, c such that $b \not\equiv c$ but $ab \equiv ac$ modulo n . Use this result to show that any coprime number has a multiplicative inverse.
5. There is a theorem by Bézout saying that given coprime a, b there are integers x, y such that $ax + by = 1$, and vice versa. Using the results of the previous two problems, show that this is true (in both directions).

2.1 Finally, Euler appears

It is a pretty well known joke between math enthusiasts that if you do not know who a theorem belongs to, it is surely either Euler or Gauss. Well, Euler's name was bound to come eventually.

There are two more big results in modular arithmetic we must cover, but the second will be covered another time. For now, we focus on using one of the results of the exercises: in particular, problem 4. For this, I greatly advise you to try it before continuing, since it spoils the main idea of the solution.

We know by the problem that given a number a coprime with the modulus, every one of its products is unique; so if we consider the multiples of a , we get all remainders again. This leads us to the same idea as with Wilson's Theorem: we multiply everything with everything and we get cancellation!

So let a be coprime with a prime number p , and consider the product

$$a \cdot 2a \cdots (p-1)a$$

This is clearly $(p-1)!a^{p-1}$; but since it's all remainders over p , it's also just $(p-1)!$. So we'd have

$$a^{p-1}(p-1)! \equiv (p-1)!$$

But by our previous result, we know $(p-1)! \equiv -1$. So we can cancel it out, and we get

$$a^{p-1} \equiv 1$$

This is huge. It tells us that we don't even care about any other number than a to find its multiplicative inverse modulo p .

The fact it only works for prime numbers is a bit annoying. Might we extend it to other numbers too?

Well, somewhat. We can use the same idea of cancelling out the multiplicative inverses, but since the multiplicative inverses are only the coprime ones, we need to consider this too. It is clear that the product of two elements with multiplicative inverses has a multiplicative inverse (*Why?*); so like before, we can multiply everything with everything *but only the elements that have multiplicative inverses*. That is, if we have $x_1, x_2, \dots, x_r$ the elements with multiplicative inverses modulo n , and we have some a coprime with n , then we consider

$$ax_1 \cdot ax_2 \cdots ax_r$$

Yet again, everything cancels out, and we get that $a^r \equiv 1$. Now this is another very strong idea: the multiplicative inverse is *always* a power of itself. This is what we saw in the Putnam problem, and why it wasn't a crazy idea to me to use a power of 3 modulo 100. But now we see this applies *everywhere*.

The only thing we don't know how to work with is this r that gives us the exponent. For now, we at least can write down in our notes the result we obtained; and we'll play around with it more later. We can, instead of writing an r , write down a function, say $\phi(n)$, that gives us our r ; just for notational convenience. We know, for example, that $\phi(p) = p - 1$ for prime numbers p . In general, we can write that, for a coprime with n :

$$a^{\phi(n)} \equiv 1$$

Now this result is gigantic, so it very much deserves a name of its own: **Fermat's Little Theorem**. The appearance of Euler's name is in the function $\phi(n)$ itself, known as Euler's totient function.

For now, we're quite good leaving it here. The introduction of these huge results is a very important step forward, so we need to make sure we understand them; and for the next time, we'll leave the last big exploration of modular arithmetic. For now, we have some more problems to solidify our knowledge and make sure everything is understood before moving forward. As before, some hints are available.

1. Calculate $\phi(6)$ (recall Euler's totient function counts all elements that have a multiplicative inverse modulo n).

2. Using a fitting value for n , use Fermat's Little Theorem to find why $x^{12} - y^4 = 2$ has no solutions for integer values of x, y .
3. Using Bézout's Theorem (given a, b and $g = \gcd(a, b)$ there are x, y such that $ax + by = g$) prove that given a coprime with n and the smallest d such that $a^d \equiv 1$ modulo n , then d divides all r that verify $a^r \equiv 1$. This d is known as the **order** of a modulo n . Is there a maximum value the order of a number can take?
4. Observe that the powers of 3 modulo 7 give all residues except 0. Considering the order of 3 modulo 7 (6) and Fermat's Little Theorem, can you argue why no number n that isn't a prime has such an a whose powers give all residues except 0 modulo n ?

2.1.1 Hints

Set 1

1. The list is $(1, 1), (2, 3), (4, 4)$
2. We can multiply numbers by $\frac{b}{b}$ and $\frac{d}{d}$; and once the fraction is the same we can group them together directly.
3. Take a common factor, or consider everything modulo g .
4. Since a is coprime with n , it has a multiplicative inverse. Using it, we can cancel out a and show that there is a contradiction. Since multiplying by a gives a unique value for each product, by size we must have some value whose product with a is 1.
5. Using problem 3, we have that if $ax + by = 1$ then $\gcd(a, b) = 1$. Using problem 4, we have that a coprime number means a multiplicative inverse; hence $ax = by + 1$ for x the multiplicative inverse of a modulo b . Moving by to the other side, we obtain Bézout's Theorem.

Set 2

1. The answer is 2 (1 and 5).
2. The modulus to use is 5. Using Fermat's Little Theorem, powers multiple of $\phi(5) = 4$ can only be 0 or 1; and a simple case work shows no combination of them gives 2.
3. Imagine d did not divide a number r such that $a^r \equiv 1$. Then, the $\gcd(d, r) = g$ is smaller than d . Using x times a^d and y times a^r , we get $a^{dx+ry} \equiv a^g \equiv 1$ since all of these products imply only 1. But since g is smaller than d , d is not the smallest, and we would have a contradiction. The order must have a cap at $\phi(n)$, since Fermat's Little Theorem forces it.
4. Since n is not prime, we know $\phi(n) < n - 1$. Thus, the powers of any a form a cycle of length at most $\phi(n) < n - 1$, so the cycle isn't long enough to cover all residues.

3 Brick by brick

When working with such an ample variety of systems that give fragments of information about objects, like modular arithmetic does, giving a small chunk of information about a number only through the lens of some other number, it is relatively sensible and sometimes useful to consider these chunks of information all together.

For example, let's take 2 and 3 at the same time. For each number x , we'll write a pair (a, b) , where $a \equiv x$ modulo 2 and $b \equiv x$ modulo 3. Like this, we can see how these moduli behave together. First things first; how many pairs (a, b) can we even have? In other words, how large is our set of possibilities? Well, since on the left all numbers have remainder 0 or 1 (2 possibilities) and on the right all have remainder 0, 1 or 2 (3 possibilities) we get a total of $2 \cdot 3 = 6$ possibilities. Further, we can see that the numbers 1 to 6 go through all possibilities:

$$\begin{array}{ll} 1 \rightarrow (1, 1) & 4 \rightarrow (0, 1) \\ 2 \rightarrow (0, 2) & 5 \rightarrow (1, 2) \\ 3 \rightarrow (1, 0) & 6 \rightarrow (0, 0) \end{array}$$

We even see that (expectedly) numbers are only $(0, 0)$ when they are divisible by 6, since they are divisible by both 2 and 3. So we can describe how numbers are divisible by 6!

Wait... didn't modulo 6 already do that? And didn't it also have size 6?

We've silently approached a very important fact. If we have the smaller parts of 6 (2 and 3) we can build 6 from them!

One could quickly jump to the conclusion that this happens to all numbers, but we can't rush here. For example, how would you break down 9? (*And do you see why it won't work?*). Clearly, $9 = 3 \cdot 3$. But if we considered 3 and 3, we can't really build a set of 9 pairs! This is obvious once you try to imagine what, for example, $(1, 2)$ would look like. So it's a number that has remainder 1 when divided by 3, but also has remainder 2 when divided by... 3. It just contradicts itself. Once again, we need to reach out to the concept of *coprime numbers*.

If the moduli we're taking are all coprime with each other (like 2 and 3) then we do, in fact, get their product.

More rigorously, we'd say that if we have coprime numbers $m_1, m_2, \dots, m_n$ then for each combination of their residues there is a corresponding residue over $M = m_1 \cdot m_2 \cdot \dots \cdot m_n$.

Even more fancily, we can just write

$$\mathbb{Z} / \prod_{i \in I} m_i \mathbb{Z} \cong \prod_{i \in I} \mathbb{Z} / m_i \mathbb{Z}$$

and leave the understanding of field theory as an exercise to the reader.

This result that allows one to treat big numbers as combinations of small ones is known as the **Chinese Remainder Theorem**.

The Chinese Remainder Theorem (CRT) gives us a lot of interesting insights into why some of the key results work, as is shown in the exercises.

1. Check that all possible combinations of residues modulo 2 and 5 appear between 0 and 9
2. What are the smallest parts a number can be broken up into that allow us to use CRT?
3. Using an argument of counting possibilities like before and CRT, prove that for coprime m, n we have $\varphi(mn) = \varphi(m)\varphi(n)$.
4. Similarly, if S_n is the number of distinct values for x^2 modulo n , prove that for coprime m, n we have $S_{mn} = S_m S_n$.

3.1 Closing remarks

Just as a little taste of the importance and applicability of modular arithmetic, I wanted to go over some interesting uses of modular arithmetic in different and interesting situations.

One might have noticed that modular arithmetic has a pretty chaotic structure; it seems even somewhat random sometimes. If, for example, one took a sequence like

$$x_{i+1} = ax_i + c \bmod n$$

Where a and c are some integers and we take an initial value x_0 , we get pretty random behaviour. With a pretty small example, given $a = 3$, $c = 2$, $n = 10$, $x_0 = 1$:

$$1 \rightarrow 5 \rightarrow 7 \rightarrow 3 \rightarrow 1$$

It seems like the sequence is rather random, and for big enough numbers it truly appears to be incredibly random, with sequences of very long periods (length) and quite chaotic behaviour.

You may have heard of the term *pseudo random number generator*; functions that make use of chaotic behaviour and a starting seed to create a random sequence of numbers for computers to use. Since a computer can't truly simulate randomness, a pseudo random number generator is the next best thing we have.

One might now see why this sequence of numbers is useful. Since it behaves mostly randomly, it ensures that from some random seed we'll get a random set of numbers to use to simulate randomness. Although this sequence isn't too

great (it fails for some randomness distribution factors) it's earned a name for its simplicity and utility: a **Linear Congruential Generator**.

One of the great things that such a pseudo random number generator has is that we ensure how large the numbers can get just by definition: all numbers are between 0 and $n - 1$. Because of this, one can set up probabilities between 0 and 1 pretty nicely, just dividing x_i by n .

Finally, for a more theoretical approach, we observe a very large applicability of modular arithmetic in terms of **Diophantine Equations**. These equations differ from others in that we focus on solving them only over the integers. For example, if we considered the following equation:

$$x^2 + y^2 = 3z^2$$

How would we go about finding integer solutions? Well, one tool would be to look at modulo 3:

$$x^2 + y^2 \equiv 0$$

Since all squares have remainder 0, or 1 modulo 3 (*verify this*) we can see that clearly

$$x \equiv y \equiv 0$$

Thus we see that $x = 3a$ and $y = 3b$ because they have remainders 0. It is clear then that

$$(3a)^2 + (3b)^2 = 3z^2 \Rightarrow 9a^2 + 9b^2 = 3z^2$$

Implies that z is also a multiple of 3 (look at it modulo 9) so we get $z = 3c$. But then we can divide out a 9

$$9a^2 + 9b^2 = 3(3c)^2 \Rightarrow 9a^2 + 9b^2 = 27c^2 \Rightarrow a^2 + b^2 = 3c^2$$

And we get a smaller solution to the same equation! So that would mean we could take any solution and divide it by 3 infinitely many times... which is only possible if $x = y = z = 0$.

Similarly, modular arithmetic has given us a taste of the types of solutions we'd have on an **perfect Euler Brick**; a prism that had all sides and diagonals (even the interior one) integer. The existence or not of such an object is still an unsolved problem in mathematics.

With this, we have reached the end of the section and the greater overall topic of modular arithmetic as a whole. But fear not! We will still observe it pop up here and there in different instances across mathematics, as all of mathematics is truly deeply connected.

3.1.1 Hints

- | | | |
|---------------------------------------|------------------------|------------------------|
| | $0 \rightarrow (0, 0)$ | $5 \rightarrow (1, 0)$ |
| | $1 \rightarrow (1, 1)$ | $6 \rightarrow (0, 1)$ |
| 1. The combinations are the following | $2 \rightarrow (0, 2)$ | $7 \rightarrow (1, 2)$ |
| | $3 \rightarrow (1, 3)$ | $8 \rightarrow (0, 3)$ |
| | $4 \rightarrow (0, 4)$ | $9 \rightarrow (1, 4)$ |
2. Since the parts must be coprime between each other, we can divide a number up to the powers of its primes. For example, $360 = 2^3 \cdot 3^2 \cdot 5$, so its smallest parts would be 5, 8 and 9.
 3. For a number to be coprime with mn , it must be coprime with both m and n individually. There are $\varphi(m)$ possible remainders coprime with m , and $\varphi(n)$ possible remainders coprime with n . Since all combinations of remainders exist, there are $\varphi(m) \cdot \varphi(n) = \varphi(mn)$ combinations of remainders coprime with mn .
 4. Clearly, each perfect square modulo mn is also a perfect square modulo m and n . Thus each combination of such perfect squares is a perfect square modulo mn (by CRT) and we get $S_m \cdot S_n = S_{mn}$.

4 Entering the imaginary

Mathematicians only strive for one goal: patterns. It is pattern finding which constitutes the basis of mathematics; taking multiple objects with some relations between them and finding properties that they share is a key factor of most theorems in mathematics (all Jordan curves have an interior and an exterior; all linear combinations of two integers are divisible by their gcd; all integers have a unique - up to order and unit multiples - prime factorisation). We'll focus on one set of objects for now: polynomials.

You might remember that a polynomial is of the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

Where each a_i is some real number. You might also recall that when polynomials are equal to 0, we call those values of x the *roots* of the polynomial. Furthermore, a polynomial can be divided by its roots. For example, if we have the polynomial

$$P(x) = x^2 - 7x + 12$$

One might find that the roots are 3 and 4. And indeed, the polynomial is the product of these values:

$$(x - 3)(x - 4) = x^2 - 7x + 12$$

One might find, however, some polynomials that seem to have no roots:

$$P(x) = x^2 - 6x + 25$$

So they are called **irreducible**.

A particularly powerful theorem regarding polynomials is known as the **Fundamental Theorem of Algebra**:

Every polynomial can be factored into parts of degree 1 or 2.

We can verify with some examples:

$$\begin{aligned}x^3 - 9x^2 + 33x - 25 &= (x - 1)(x^2 - 8x + 25) \\x^3 - 6x^2 + 11x - 6 &= (x - 1)(x - 2)(x - 3) \\x^4 + 3x^2 + 2 &= (x^2 + 1)(x^2 + 2)\end{aligned}$$

This makes it clear that, essentially, the key polynomials are the quadratic and linear ones. We can think of them like the primes of polynomials, the atoms of their world.

There is an issue that arises in this theorem, however.

If one plays around with sets of random polynomials, it might be hard to exactly determine just from the polynomial if or not it has irreducible quadratic parts. And when working with very general objects we never like them having differences in such a fundamental point.

This is where our protagonist finally becomes the key. By introducing i , we can forget about all of our problems.

You might recall the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

And even remember the discriminant:

$$\Delta = b^2 - 4ac$$

Together with a strict categorization of its values: if it's positive, the polynomial has 2 roots; if it's 0, the polynomial has 1 (doubled) root; if it's negative, we have no roots.

Here is the key. If we could always have 2 roots, all issues would be fixed; any quadratic polynomial could be factored into linear (degree 1) polynomials. But if Δ is negative, we would be taking the square root of a negative number, which is wrong... right?

One of the greatest revolutions in mathematics was breaking free from the bounds of reality. If there is no value that squares to a negative number, then let's just come up with it!

Mind you, this was controversial. Yet somehow, although it was wrong from a real point of view, adding in i made a lot of issues just go away. All polynomials factored into degree 1 and 2? Silly. **All polynomials can be factored into degree 1.** Geometry. Trigonometry. Physics! The very language of reality!

Although they started their journey as a simple fix in polynomials, complex numbers are now a pillar of modern mathematics.

So, what are they anyway?

A complex number is any number of the form

$$a + bi$$

Where a, b are real numbers, and $i = \sqrt{-1}$. You might wonder how it's even possible to add a real number to an imaginary (multiple of i) number in this way; imaginary numbers don't have decimal representations! But here we have

to accept that imaginary numbers *shouldn't* have decimal representations. The number $a + bi$ is a single number, we just don't (and can't) represent it in any decimal representation.

Think about it like if you had $\pi + 1$. This is one single number, but since π is an infinite decimal number we simply write the Greek letter in its place. And adding 1 to it, we still obtain a specific number, but writing it using this sum is more useful and conveys more meaning. In the same way, seeing the two parts of a complex number gives us information about it.

Complex numbers add and multiply in a natural way:

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

$$(a + bi)(c + di) = ac + adi + bci + bdi^2 = (ac - bd) + (ad + bc)i$$

Now, finally, we can begin understanding the power behind complex numbers.

We are all familiar with the real line:

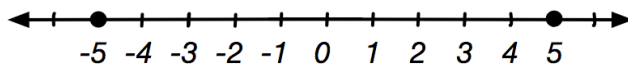


Figure 1: The real line

But where in the line do we place i ? Well, this requires a quite out-of-the-box thought, but we don't! After all, since i is imaginary, we could say it's *above* the real numbers. Thus, we instead make the **complex plane**:

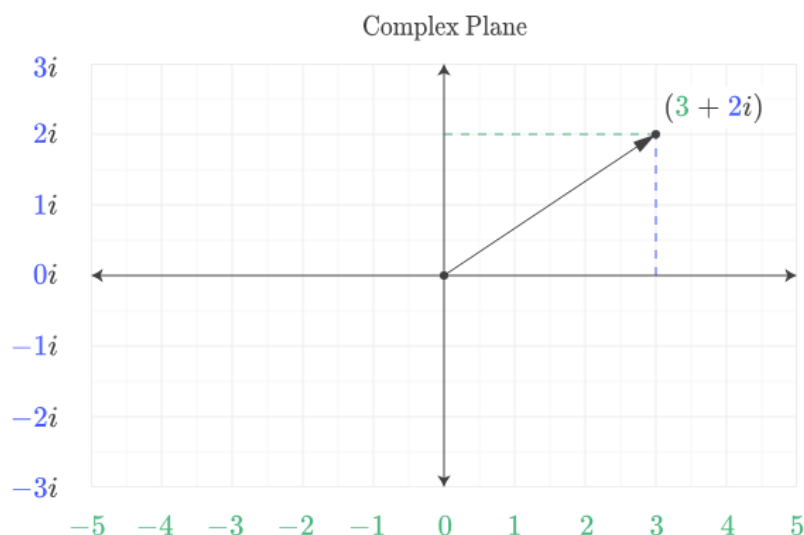


Figure 2: The complex plane

As is clearly visible, thinking of the complex plane allows us to visualise complex numbers as vectors or points in the complex plane. And as is clear from their addition, they do add (and subtract) just like vectors.

Now, although they act like vectors, they're not vectors. Because they also have the power of multiplication, which brings in a very powerful tool to the table.

Before that, we'll take a quick break to ensure that we understand the complex numbers and complex plane:

1. Using the Pythagorean Theorem, calculate how long the vector in figure 2 is.
2. Verify that $3 + 4i$ and $3 - 4i$ are roots of $x^2 - 6x + 25$. Furthermore, verify that the quadratic formula returns these values.
3. Considering that a real number has an imaginary part of 0, find that if $(a + bi)(c + di) = r$ where r is real then $ad = -bc$, or, equivalently, $\frac{a}{b} = -\frac{c}{d}$.
4. Using the previous result, find that the only complex number that both adds and multiplies to a real number with $a + bi$ is $a - bi$. This is called the **complex conjugate** of a complex number.

4.1 Turn around

Sadly, to understand the power of multiplication of complex numbers, we first need to recall a bit of trigonometry.

You might recall the soh-cah-toa mnemonic: sine is opposite over hypotenuse, cosine is adjacent over hypotenuse, and tangent is opposite over adjacent. Since the complex plane divides the real and imaginary parts into vertical and horizontal with a right angle, we can rewrite any complex number as

$$a + bi = r(\cos \theta + i \sin \theta)$$

Where r is the length of the vector, and θ is the angle it forms with the x-axis. Also, we write θ not in degrees, but in radians.

Radians are simply a more mathematically useful way to write angles. Instead of 360° representing a full turn, we represent a full turn with 2π radians. Everything else works the same; half a turn is π radians, a quarter is $\frac{\pi}{2}$... You should think of it like Fahrenheit and Celsius: two ways of saying the same.

It turns out that if we use angles in radians, we find that there is a particularly beautiful identity that holds (the details of which we skip for reasons any experienced mathematician could tell); **Euler's Identity**:

$$\cos \theta + i \sin \theta = e^{i\theta}$$

Thus, any complex number could be written as

$$a + bi = re^{i\theta}$$

Here lies the power of complex numbers. If we had two complex numbers $a + bi$ and $c + di$, we could write their product as

$$(a + bi)(c + di) = (r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = (r_1 r_2) e^{i(\theta_1 + \theta_2)}$$

Which in fact is a complex number of length $r_1 r_2$ and angle $\theta_1 + \theta_2$! Thus, **a complex number product is a scaling and rotation**.

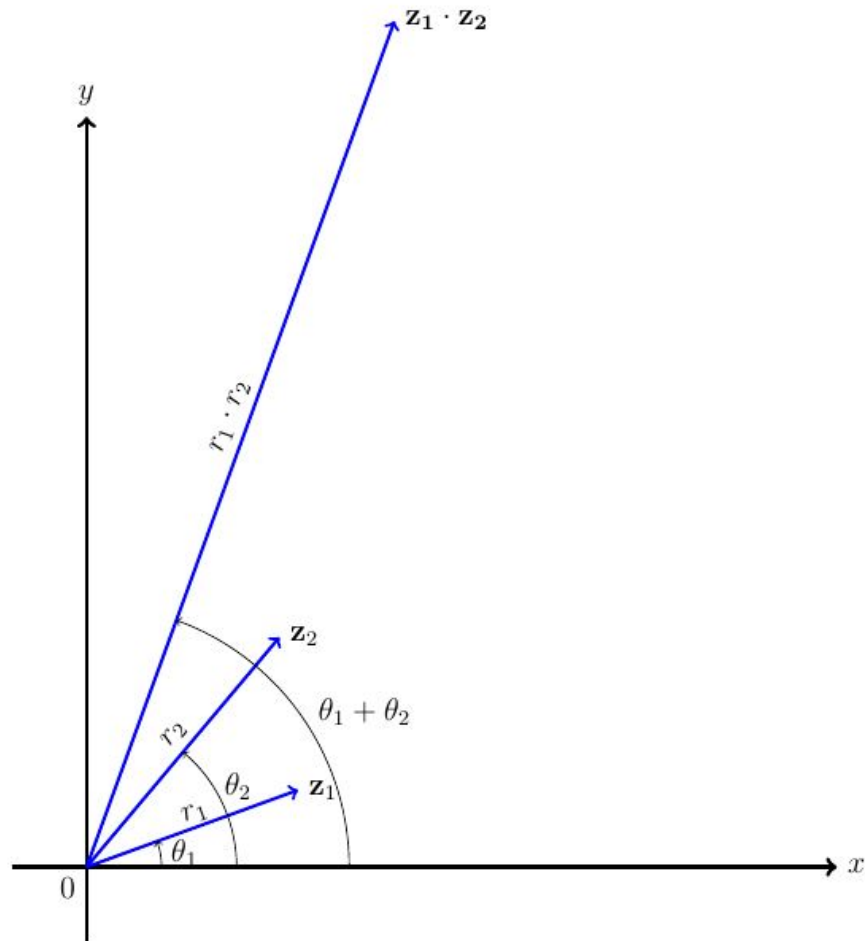


Figure 3: Product of two complex numbers

For now, we must leave this section here. Sadly, it had to be more on the raw explanations side of things; but as is usual with mathematics, we still have an infinity to explore!

We leave a couple problems to finish off and strengthen the understanding of these particularly fascinating numbers.

1. We have a hexagon in the complex plane with its center at 0, and one of its vertices at 2. Where are the remaining vertices placed?
2. Using the radius-angle form (called the *modulus-argument* form) find that for any complex number $a + bi$ (nonzero) there is a *multiplicative inverse* (recall that the multiplicative inverse of x is the number x' such that $xx' = 1$).
3. Using an angle argument, argue why only the scalar multiples (i. e. multiplied by a real number) of the complex conjugate of a number can multiply to a real number with it (for the definition of complex conjugate, go to exercise 4 of section 1).
4. Is the one piece real?

4.1.1 Hints

Set 1

1. It would be of length $\sqrt{3^2 + 2^2} = \sqrt{13}$.
2. We can check that they are roots directly:

$$(3 + 4i)^2 - 6(3 + 4i) + 25 = 9 + 24i - 16 - 18 - 24i + 25 = 0$$

$$(3 - 4i)^2 - 6(3 - 4i) + 25 = 9 - 24i - 16 - 18 + 24i + 25 = 0$$

As for the quadratic formula, directly substituting we find:

$$x = \frac{-(-6) \pm \sqrt{36 - 4(1)(25)}}{2(1)} = \frac{6 \pm \sqrt{-64}}{2} = 3 \pm 4i$$

3. Simply expanding: $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$. Since r is real, $ad + bc = 0 \Rightarrow ad = -bc$.
4. For a number to add to a real number with $a + bi$, it must have imaginary part $-bi$. Using that if it multiplies to a real number then $a(-b) = -bc$, we have $c = a$, so $a - bi$ is the only solution.

Set 2

1. Using that the angle between the center of the hexagon and the vertices are all 60° , we can use that 60° is $\frac{\pi}{3}$ radians to find $e^{i\frac{\pi}{3}} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$. We multiply this by 2 5 times to find each of the remaining vertices: $1 + i\sqrt{3}, -1 + i\sqrt{3}, -2, -1 - i\sqrt{3}, 1 - i\sqrt{3}$.
2. If $a + bi = re^{i\theta}$, the number $\frac{1}{r}e^{i(-\theta)}$ will be its multiplicative inverse; the radii multiply to 1, and the angles add to 0.
3. It is visually clear that the complex conjugate and its multiples always have an angle of $-\theta$. In fact, it is clear all complex numbers of angle $-\theta$ are scalar multiples of the complex conjugate. Thus, for the angles to cancel out, we need the complex conjugate or one of its multiples. For a different way to arrive to the same, we can also observe that the condition $\frac{a}{b} = -\frac{c}{d}$ from exercise 3 of set 1 is in fact a condition for the tangent of the angle, which leads us again back to $-\theta$.
4. No.

5 Imaginary numbers in real time

This section of Imaginary Numbers was carried out in a call as a lesson, with interactive elements for my "students" (friends under coercion). The resources can be found at:

<https://www.geogebra.org/m/ne7bsmc6>

<https://www.geogebra.org/m/gphrakkk>

6 A small show of modular arithmetic at play

I recently stumbled across an interesting problem that shows pretty well some of the strengths of modular arithmetic and prime number arguments:

For a given positive integer m , find all triples (n, x, y) of positive integers, with n relatively prime (coprime) to m , which satisfy $(x^2 + y^2)^m = (xy)^n$

To solve this, we need to consider a very particular modulus: let p be a prime number that divides y but not x . Considering the equation modulo p :

$$\begin{aligned}(x^2 + 0^2)^m &\equiv (x \cdot 0)^n \pmod{p} \\ x^{2m} &\equiv 0 \pmod{p}\end{aligned}$$

Which can only be true if $x \equiv 0 \pmod{p}$, therefore there are no prime numbers that divide y but not x . Since x and y are symmetric, we can see that the same is true in reverse. Thus, since they share every factor, $x = y$. We thus now have

$$\begin{aligned}(2x^2)^m &= (x^2)^n \\ 2^m x^{2m} &= x^{2n}\end{aligned}$$

Observe now that if p^r is an odd prime number power that divides x , we have that $p^{2mr} = p^{2nr}$, which contradicts m, n coprime (and the case $m = n = 1$ is clearly broken). Thus, $x = 2^r$ since no odd prime number divides it. Thus

$$\begin{aligned}2^m \cdot 2^{2mr} &= 2^{2nr} \\ 2^{m+2mr} &= 2^{2nr} \\ m(2r + 1) &= 2nr\end{aligned}$$

Since m, n are coprime, and $2r, 2r + 1$ are also coprime, they correspond to each other:

$$m = 2r, n = 2r + 1$$

So the solutions are the following:

- For odd m , there are no solutions.
- For even m , we have $n = m + 1$, and $x = y = 2^{\frac{m}{2}}$.

□

7 Olympiad mathematics: Algebra

Finally, we begin with the (fun) torture of olympiad mathematics in its full form. This time, we'll start by looking at algebra, which will include the exciting topics of:

1. Sequences
2. Inequalities
3. Functional equations
4. Summations

But do not fear; all of these will be more than reasonable in depth with many little examples and some problems that will be left for you to try out (without solutions, so sending in your own proofs and asking in case of doubt is encouraged!). The problems at the end will be more or less in order of difficulty, and particularly hard problems will be marked with an asterisk as is typical. Nevertheless, I greatly encourage you to not fear the asterisks; whoever collects more stars (asterisks) gets a prize! (this is not a contractual agreement and the prize is TBD :3).

So let's begin!

7.1 Sequences

Most of us have worked with at least the basic sequences, which we might recognize:

- The **arithmetic sequence**, where consecutive numbers have constant difference:

$$a_{n+1} = a_n + d$$

For example: 1, 4, 7, 10, 13, ... and 2, 4, 6, 8, 10,

- The **geometric sequence**, where consecutive numbers have constant quotient:

$$a_{n+1} = a_n r$$

For example: 1, 3, 9, 27, 81, ... and 5, 10, 20, 40, 80,

We will go over these sequences, but will also explore a world of other sequences, and learn methods to find facts and conclusions about them. As a sneak peek, we can look at what we'll find in sequence techniques:

1. MSAP, PSIP, RP
2. Induction
3. Invariants
4. Limits

5. Bounds and monotone sequences

6. Generating functions

Each of these methods leads to deeper understandings of sequence work. One should think of these as tools on a belt; the more, the better we can tackle various problems and find conclusions on them.

Before plunging in completely, we ought to introduce some of the important terminology we'll use in the context of sequences. Firstly, the **general term** of a sequence is just a description of a general value of the sequence without depending on other values or weird sums; we can also call it a **closed form**. For example, for the geometric sequence, the general term would be $a_n = a_1 \cdot r^{n-1}$, since this doesn't depend on changing values (it does depend on a previous value, a_1 , but this is always *constant*, so we don't care too much) except for a value we can calculate with one computation (r^n).

We also have the **recursive definition** of a sequence; a description of a value that depends on previous values, like the arithmetic and geometric sequences as defined above or the Fibonacci sequence, $F_{n+2} = F_{n+1} + F_n$.

And with these few bits of terminology, we're ready to plunge right in!

7.1.1 MSAP, PSIP, RP

In this section we discuss the arguably most important tools, and at that the simplest ones. (Also, do not cite these principles in any official competition; they're purely of my invention so the point goes across easily and clearly). Firstly, we discuss the **Move Shit Around Principle**.

In any situation of sequence work, a lot of sequences include recursions, conditions and constraints of some sort. Our first and key idea is to very simply try playing around with the conditions to obtain more information. Let us consider first the infinite geometric sum. Recall that a geometric sequence has the form:

$$a_n + 1 = a_n r$$

For example, if $a_1 = 2$ and $r = 3$, we get:

$$a_2 = a_1 \cdot 3 = 2 \cdot 3 = 6$$

$$a_3 = a_2 \cdot 3 = 6 \cdot 3 = 18$$

$$a_4 = a_3 \cdot 3 = 18 \cdot 3 = 54$$

So in general the sequence looks like:

$$2, 6, 18, 54, \dots$$

And the general term is $a_n = a_1 \cdot r^{n-1}$.

Now, if $r < 1$, we can consider the infinite sum of the terms:

$$\begin{aligned} S_\infty &= a_1 + a_2 + a_3 + \cdots \\ S_\infty &= a_1 + a_1 r + a_1 r^2 + \cdots \end{aligned}$$

Now, if we apply the **Move Shit Around Principle**, we want to try doing stuff to this sum. For example, one could see that all terms after a_1 are multiples of r , so we can factor an r out:

$$S_\infty = a_1 + r(a_1 + a_1 r + a_1 r^2 + \cdots)$$

But wait... the terms inside the parentheses are exactly our sum again! So we can substitute it in:

$$\begin{aligned} S_\infty &= a_1 + r(S_\infty) \\ S_\infty - rS_\infty &= a_1 \\ S_\infty(1 - r) &= a_1 \\ S_\infty &= \frac{a_1}{1 - r} \end{aligned}$$

And we obtain the well known sum formula, which one can even find used in our previous complex number lecture.

The main idea with these principles is usually to just play around with the sequence until some new information is obtained, or a better way of writing the recursions is found. Another example would be:

$$a_{n+2} = 2a_{n+1} - a_n$$

Using again the move shit around principle, we see that there's a very simple way to get a pretty symmetric recursion:

$$a_{n+2} - a_{n+1} = a_{n+1} - a_n$$

So the difference between consecutive terms is always constant. Say we label it d :

$$\begin{aligned} a_{n+1} - a_n &= d \\ a_{n+1} &= a_n + d \end{aligned}$$

Wait... this is just an arithmetic sequence.

So by just moving shit around, we found some very nice conclusions about the sequence itself, without applying any of the tools that we'll see further. This is why we should first and foremost apply these principles, and then attempt to use our more powerful (and sometimes overkill) tools. Now, we move on to a next problem that'll show us insights into the other two key features.

Problem. Find the general term of the sequence

$$a_{m+n} + a_{m-n} = \frac{1}{2}(a_{2m} + a_{2n})$$

For $m \geq n \geq 0$, and such that $a_1 = 1$.

Proof. We use the two other techniques. Firstly, we use the **Plug Shit In Principle**. This principle states that we should try plugging in some nice, simple values into the sequence. For example, we could try the simplest values for starters:

$$m = n = 0$$

$$a_0 + a_0 = \frac{1}{2}(a_0 + a_0)$$

$$a_0 = 2a_0 \Rightarrow a_0 = 0$$

$$m, n = 0$$

$$a_m + a_m = \frac{1}{2}(a_{2m} + a_0)$$

$$4a_m = a_{2m}$$

$$m, n = 1$$

$$a_{m+1} + a_{m-1} = \frac{1}{2}(a_{2m} + a_2)$$

$$a_2 = 4a_1 = 4 \text{ (by our previous result)}$$

$$a_{m+1} + a_{m-1} = \frac{1}{2}(4a_m + 4) = 2a_m + 2$$

The a_{m+1} and $2a_m$ could remind you of what we did previously, and we might try to obtain a symmetric equality once again:

$$a_{m+1} - a_m = a_m - a_{m-1} + 2$$

Now, although this looks so temptingly close to what we had before, that 2 doesn't quite let us have what we want. However, we can now use the **Rephrase Principle**: whenever you have somewhat symmetric setups, it might be useful to find a new sequence that is formed from the previous one to find a better phrasing of the problem. This can become clearer with our example. Since

on the left and right we have the same symmetric setups, we can create a new sequence $(b_n)_{n \geq 0}$ defined as

$$b_n = a_{n+1} - a_n$$

And with that, we get our previous result rephrased as

$$b_n = b_{n-1} + 2$$

Which is just an arithmetic sequence! That, we know how to work with. We can now see that if we add together the b_n , the positive and negative terms cancel (this is a clear depiction of telescoping, which we cover in the sums section). Therefore:

$$\begin{aligned} a_{n+1} &= a_{n+1} + (-a_n + a_n) + (-a_{n-1} + a_{n-1}) + \cdots + (-a_0 + a_0) \\ a_{n+1} &= (a_{n+1} - a_n) + (a_n - a_{n-1}) + \cdots + (a_1 - a_0) + a_0 \\ a_{n+1} &= b_n + b_{n-1} + \cdots + b_0 + a_0 \end{aligned}$$

And to solve that right side we'll simply use the standard formula for an arithmetic sequence sum (which, not to worry, we'll prove in the next section):

$$a_{n+1} = \left(1 \cdot (n+1) + 2 \frac{n(n+1)}{2} \right) + 0 = n+1 + n^2 + n = n^2 + 2n + 1 = (n+1)^2$$

And so we find that the sequence is in fact the perfect squares sequence:

$$0, 1, 4, 9, 16, 25, \dots$$

Hopefully, these few examples have shown the power that the MSAP, PSIP and RP have over sequences; most problems rely on just playing around with these three tools, and perhaps later adding on the other tools one learns.

To close off the section, we leave some practice problems to apply these basic principles. These problems *do not* include hints. We instead highly encourage trying them without any help first, and asking for any hints and proof checking directly.

1. Find the general term of the sequence

$$a_{n+2} = \frac{a_{n+1}^2}{a_n}$$

2. Find a recursive relation for the sequence

$$2, 2, 4, 6, 10, 16, 26, 42, 68, \dots$$

3. * Using a telescoping argument, find the sum of the first n terms of the sequence defined by

$$a_{n+2} = 3a_{n+1} - a_n$$

in terms of only the values of a_{n+1}, a_n, a_1, a_0

4. * Knowing $r \cdot r^2 \cdot r^3 \dots r^n = r^{\frac{n(n+1)}{2}}$, find the general term of the sequence

$$a_{n+2} = \frac{a_{n+1}^2}{a_n} r$$

with $a_1 = 1$

8 Calc I

8.1 Continuity

To define continuity, we try to formalise the usual notion:

"A continuous function is that which you can draw without raising your pen."

Now, what constitutes a "drawable" function? Well, we could say that there are no "jumps", that is, no place where the function breaks from one point to another. This defines a couple of possible "bad" situations:

1. **A missing point.** If we find that a function is literally missing a point, we cannot possibly draw it without raising our pen, even if for just one point.
2. **A stop and continuation.** If the function gets to a point, and then continues from a completely different point, we must raise our pen from one to the other.
3. **An infinite jump.** This is like the stop and continue, but in the infinite case. If the function just keeps going to ∞ and then comes back from $-\infty$, we must raise our pen, even if at no point we "stopped", per se.

Now, all of these have already been classified, and have handy names to refer to them. We can see them in figure 4.

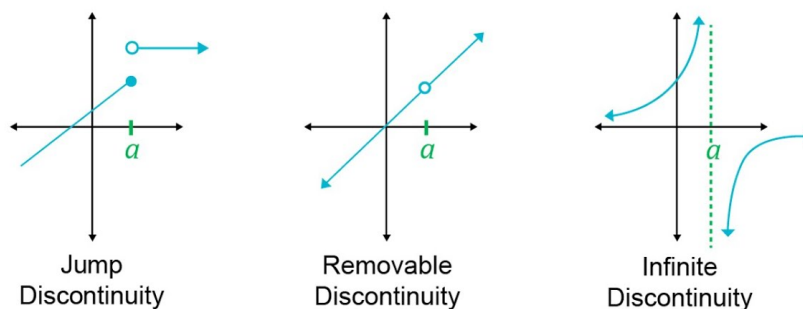


Figure 4: Types of discontinuities

We must tackle these in two ways. For the removable discontinuity, we'll simply

have to limit continuity to sections of a function that *don't* have such discontinuities. Thus, we say that a function f is *continuous in the segment* (a, b) , where we require all points in (a, b) to be defined in f . It's the simple solution, but it works.

Now, for the jump/infinite discontinuities, we'll use a clever phrasing of this problem. We can say that, at the point x_0 where this discontinuity is found, we can move just a little in one direction (left or right) and immediately jump from $f(x_0)$ to a completely different $f(x_0) + \varepsilon$ (observe the first image in 4).

So, to prove that a point x_0 is continuous, we want to prove the opposite: that if we move a bit to the left or right, we don't suddenly jump a big amount. More technically, we want that if from x_0 we check another value close by, say $x_0 + \delta$, that $\varepsilon = |f(x_0 + \delta) - f(x_0)|$ isn't too big. In fact, if δ gets smaller, we want ε to get smaller. In fact!! we say that as δ goes to 0, that ε also goes to 0.

Let's see it in a figure (figure 5):

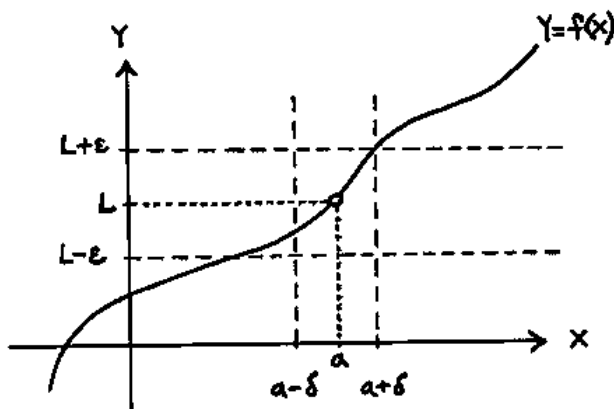


Figure 5: Epsilon-Delta definition of continuity

We can see that, as δ gets smaller, the vertical lines get closer to the point a , and then the horizontal lines also get closer, so ε gets smaller. Because of the variables, this is usually referred to as the **Epsilon-Delta definition of continuity**. Does this mean there are other definitions? Yes! But they're defined in other fields of math; for example, there is the **Open Set** definition, but it pertains to *Topology*, which we'll cautiously avoid for now.

Now, we know math is about formulas, so let's set up a formula for this. We said that we want to see how, as $\delta \rightarrow 0$, $\varepsilon \rightarrow 0$. This sounds like the limit:

$$\lim_{\delta \rightarrow 0} \varepsilon = 0$$

In which we can substitute our definition of ε to get:

$$\lim_{\delta \rightarrow 0} |f(x_0 + \delta) - f(x_0)| = 0$$

And that's it! If we prove that at all points in (a, b) the function verifies this formula, we can get continuity in (a, b) .

8.1.1 A simple example

Let's see how $f(x) = x$ is continuous in $(0, 1)$, for example.

First, we have to choose a point $x_0 \in (0, 1)$. Perhaps we can choose $x_0 = \frac{1}{2}$, for example. Well, now we look at the limit:

$$\lim_{\delta \rightarrow 0} \varepsilon = 0$$

Which, as we said, is:

$$\lim_{\delta \rightarrow 0} |f(x_0 + \delta) - f(x_0)| = 0$$

So let's substitute all we can:

$$\begin{aligned} \lim_{\delta \rightarrow 0} \left| \left(\frac{1}{2} + \delta \right) - \left(\frac{1}{2} \right) \right| &= \\ &= \lim_{\delta \rightarrow 0} |\delta| \end{aligned}$$

Which, by definition, does give 0! Thus, at $\frac{1}{2}$, the function $f(x) = x$ is continuous.

Equipped with this, we can thus see how to *disprove* continuity.

8.1.2 The Dirichlet function

The Dirichlet function is the most classical example of a *nowhere continuous* function. That is, that no matter what x_0 we choose, it *never* verifies our formula!

The function is defined as follows:

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$$

How would one prove this is nowhere continuous? Well, let's choose a specific point.

Let's say we take $x_0 = 0$. This is a rational number, so $f(0) = 1$. Now, we look at the limit:

$$\lim_{\delta \rightarrow 0} \varepsilon = 0$$

$$\lim_{\delta \rightarrow 0} |f(x_0 + \delta) - f(x_0)| = 0$$

Substituting:

$$\lim_{\delta \rightarrow 0} |f(\delta) - 1| = 0$$

For this to be true, we would require that $f(\delta) \rightarrow 1$, obviously. But now we use a clever idea:

Lemma 8.1 *When doing a limit, we can choose any path to arrive to the limit.*

That is, in our example, the path to 0 could be $\delta = 1$, $\delta = 0.1$, $\delta = 0.01$, $\delta = 0.001$, ...

Or, alternatively, it could be $\delta = 1$, $\delta = \frac{1}{2}$, $\delta = \frac{1}{3}$, $\delta = \frac{1}{4}$, $\delta = \frac{1}{5}$, ...

Or, which is the one we'll use, it'll be $\delta = \frac{1}{\pi}$, $\delta = \frac{1}{2\pi}$, $\delta = \frac{1}{3\pi}$, $\delta = \frac{1}{4\pi}$, ...

We can see all of these do approach 0, although through different paths. Nonetheless, any limit allows us to choose *any* path, as long as it approaches the limiting value.

Taking the third path, $\delta = \frac{1}{\pi}$, $\delta = \frac{1}{2\pi}$, $\delta = \frac{1}{3\pi}$, $\delta = \frac{1}{4\pi}$, ..., we see all of these are **irrational** numbers. Thus, $f(\delta) = 0$ at every point of the sequence. Which, in our formula, means:

$$\begin{aligned} \lim_{\delta \rightarrow 0} |f(\delta) - 1| &= \\ &= \lim_{\delta \rightarrow 0} |0 - 1| = \\ &= 1 \neq 0 \end{aligned}$$

And that's it! At $x_0 = 0$, the function is *not* continuous. Could this logic be applied at every point? Well, yes! As long as the path guarantees the value $x_0 + \delta$ stays the opposite of x_0 (rational/irrational), the difference is 1, so the function is *discontinuous*.